

CBSE Practice Worksheet

Class 7 | Science | Difficulty: Hard | Set B

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
2. Which quantum number determines the shape of an orbital?
(A) Principal
(B) Azimuthal
(C) Magnetic
(D) Spin
3. Which organelle is involved in programmed cell death?
(A) Lysosome
(B) Mitochondria
(C) Peroxisome
(D) Ribosome
4. What is the efficiency of an ideal Carnot engine operating between 300K and 400K?
(A) 25%
(B) 33%
(C) 50%
(D) 75%
5. What is the escape velocity from Earth's surface?
(A) 8 km/s
(B) 11.2 km/s
(C) 15 km/s
(D) 25 km/s
6. Which type of mutation involves change in single nucleotide?
(A) Deletion
(B) Insertion
(C) Point mutation
(D) Frameshift
7. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
8. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis

(D) Respiration

9. What is the oxidation state of Mn in KMnO_4 ?

- (A) +2
- (B) +4
- (C) +6
- (D) +7

10. What is the wavelength of Lyman series limit?

- (A) 91.2 nm
- (B) 121.6 nm
- (C) 656.3 nm
- (D) 364.6 nm

Section B: Fill in the Blanks (1 mark each)

- 11. The uncertainty principle was given by ____.
- 12. The enzyme that converts prothrombin to thrombin is ____.
- 13. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 14. The process of splitting water during photosynthesis is called ____.
- 15. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.
- 16. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 17. The band gap in semiconductors is typically ____ eV.
- 18. The Krebs cycle occurs in the ____ of mitochondria.

Section C: Short Answer Questions (2 marks each)

- 19. Derive the lens formula and explain its applications.
- 20. Describe the light-dependent and light-independent reactions of photosynthesis.
- 21. Describe the mechanism of nerve impulse transmission.
- 22. Explain the structure and working of a nuclear reactor.
- 23. Explain Le Chatelier's principle with examples.
- 24. Derive the equation for kinetic energy of gases from kinetic theory.
- 25. Explain the mechanism of DNA replication.

--- End of Worksheet ---

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